

SAUMYA AGRAHARI

Software Development Engineer | Full Stack Developer | AI/ML Engineer | Python Developer

+91-8318286501 saumyaagrahari262730@gmail.com Prayagraj, Uttar Pradesh [linkedin.com/in/saumya-agrahari](https://www.linkedin.com/in/saumya-agrahari) github.com/saumya3005

Taylor & Francis Group | 3x Hackathon Winner (UHack 4.0 #1/500+ Participants, Google Build with AI GDG, NSUT Top 10)

Professional Summary

Final-year B.Tech (Artificial Intelligence & Machine Learning) student with hands-on experience through internships, freelance development, research, and production-grade software projects. Published researcher with Taylor & Francis Group and 3x hackathon winner, experienced in building scalable REST APIs, AI/ML pipelines, and full-stack applications using Python, React.js, Node.js, Django, TensorFlow, Docker, and cloud deployment technologies. Strong foundation in System Design, Agile/Scrum, and SDLC. Seeking Software Development Engineer, Full Stack Developer, and AI/ML Engineer opportunities.

Education

B.Tech — Artificial Intelligence & Machine Learning

United University, Prayagraj

Aug 2023 – May 2027

CGPA: 8.4/10 | SGPA: 9.0

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, DSA, DBMS, Operating Systems, Computer Networks, System Design, Agile Software Development

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C

Frontend: React.js, Next.js, Redux, Context API, HTML5, CSS3, Tailwind CSS, Bootstrap, Responsive Design

Backend: Node.js, Django, Flask, REST API Design, JWT Authentication, API Integration, Microservices

Databases: MySQL, PostgreSQL, Firebase Firestore, Database Design, Query Optimization

AI/ML: TensorFlow, Scikit-learn, XGBoost, OpenCV, MediaPipe, Computer Vision, NLP, NumPy, Pandas, Feature Engineering, Model Deployment

Generative AI: Google Gemini API, LLM Integration, Prompt Engineering, Vertex AI

Cloud & DevOps: AWS (S3, EC2), GCP, GitHub Actions (CI/CD), Docker, Git, Vercel, Linux

Testing: Jest, Pytest, Unit Testing, API Testing, Test-Driven Development (TDD)

Core CS: DSA, OOP, System Design, SDLC, Agile/Scrum, Software Architecture

Work Experience

Prodesk IT, Noida

Backend Development Intern

Aug 2025 – Present

- Engineered 10+ REST API endpoints in Python/Django for R&D production applications, improving data retrieval by 25% via query optimization and indexing strategies.
- Wrote Pytest unit tests for 3 core API modules, achieving 85%+ test coverage; designed scalable MySQL database workflows reducing backend response time by 30%.
- Collaborated in Agile/Scrum team of 6+ engineers across 3 production releases; participated in daily standups, sprint planning, and code reviews.

Code Resite, Prayagraj

Web Developer Intern

Jun 2025 – Jul 2025

- Delivered full-stack features across 5+ client projects using React.js, JavaScript, Node.js, and REST APIs; reduced UI bugs by 40% via systematic testing and debugging.
- Improved page load performance by 35% through frontend optimization (lazy loading, code splitting, caching); all projects deployed on Vercel with CI/CD pipelines.

CodeAlpha

Full Stack Development Intern

Jun 2026 – Jul 2026

- Developed responsive full-stack web applications using React.js, JavaScript, HTML5, CSS3, and REST API integration as part of the Full Stack Development Internship Program.
- Implemented authentication, CRUD operations, reusable UI components, and responsive layouts while following modern web development best practices.
- Collaborated in an Agile development environment, maintained source code using Git/GitHub, and successfully completed all assigned development tasks within the internship duration.

Hanuman Pushpavarsha Committee, Prayagraj

Freelance Full Stack Developer

Jan 2024 – Mar 2025

- Built bilingual (Hindi-English) production website using Next.js, Firebase, and Vercel with 6 modules: Razorpay payment gateway, live streaming, member registry, admin dashboard; serving 500+ users.
- Reduced admin overhead by 60% by automating member registration via Firebase Firestore real-time database; implemented Firebase Auth with role-based access control (RBAC).

Projects

SignSetu — AI Bidirectional Sign Language Platform | Python, TensorFlow, OpenCV, MediaPipe, React.js, Flask, Firebase, Docker Live | [GitHub](#)

- Winner, UHack 4.0** (1st/500+ participants): real-time bidirectional ISL communication platform for the deaf and hard-of-hearing community; deployed with Docker and Vercel CI/CD.
- Trained TensorFlow CNN + MediaPipe achieving **91%+ gesture recognition at 24 fps**; curated 5,000+ sample custom ISL dataset from scratch using feature engineering and data augmentation.

- Architected Sign-to-Text, Sign-to-Speech, and Text-to-Sign (3D avatar) pipelines with JWT authentication, Firebase backend, and REST API endpoints.

Vynk — Full-Stack Social Networking Platform | *React.js, Node.js, Express.js, MongoDB, JWT, Tailwind CSS, Socket.io, REST APIs*
[Live](#) | [GitHub](#)

- Developed a full-stack social networking platform with secure JWT authentication, user profiles, post creation, likes, comments, media sharing, and personalized feeds.
- Engineered scalable RESTful APIs using Express.js and MongoDB with modular backend architecture, role-based authentication, and optimized CRUD operations.
- Built a responsive React.js interface with Tailwind CSS and integrated real-time interactions using Socket.io to enable instant notifications, messaging, and seamless user engagement.

FlowSync — Collaborative Project Management Platform | *React.js, Node.js, Express.js, MongoDB, JWT, Tailwind CSS, REST APIs*
[Live](#) | [GitHub](#)

- Designed and developed a Trello/Asana-inspired project management platform supporting project creation, task assignment, progress tracking, deadlines, and team collaboration.
- Implemented secure JWT authentication, project boards, task management, comments, and role-based access using Express.js, MongoDB, and RESTful APIs.
- Built an interactive dashboard with reusable React components and responsive UI, enabling efficient project workflows and improved team productivity.

Research Publication

Multi-Class Anomaly Detection in Network Traffic Using Supervised Machine Learning | [DOI](#)

Saumya Agrahari, Ayush Tiwari, Priyanka | **Taylor & Francis Group Computer Science** | ICMLDE 2025 | AICCT-2025, Sept 2025

- ML-based IDS on CIC UNSW-NB15 (175,000+ records); benchmarked 4 models (Decision Tree, Random Forest, Logistic Regression, XGBoost); **93.78% accuracy with Precision, Recall, F1, ROC-AUC**. Selected for international conference presentation.

Awards & Leadership

- **Winner, UHack 4.0:** SignSetu — 1st/500+ participants, real-time ISL AI platform.
- **Winner, Google Build with AI (GDG):** Developed an AI-powered solution recognized by Google Developer Group ecosystem judges for innovation and practical impact.
- **Top 10 Finalist, NSUT National Hackathon, New Delhi:** National-level competitive hackathon.
- **Lead Organizer, HackDiwas 3.0:** 500+ participant state-level hackathon; **TEDx United University Organizer:** speaker outreach and technical operations.
- **Community:** GDG Prayagraj , Wikiclub tech , Prerogative pointers 2025 , FOSS Prayagraj .